ASSESSMENT BRIEF

Part 1: Applied problem-solving activity

Programme	MSc Data Science		
Module Title	Mathematics for Data Science		
Module Code	CMP-L011-0		
Module Level	7	**Assessment Type(s)**	Report
Word Length / Duration	1000 words	**% contribution to module mark**	40%
Deadline (date & time) for Submission	10th March 2026 (Friday)	**Format/Location of submission**	Moodle
Assessment Feedback date:	17th March 2026 (Friday)		

Learning Outcomes

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below.

By successfully completing this assessment, you will be able to

- Apply linear algebra operations (vectors, matrices) to real-world data representations.
- Perform descriptive statistics and exploratory data analysis (EDA) with mathematical justification.
- Formulate and test hypotheses using inferential statistics.

Employability and Professional Skills

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs.

By successfully completing this assessment, you will be able to

- Apply linear algebra operations (vectors, matrices) to real-world data representations.
- Perform descriptive statistics and exploratory data analysis (EDA) with mathematical justification.
- Formulate and test hypotheses using inferential statistics.

• Model data with probability functions (discrete and continuous) and interpret results. • Critically reflect on mathematical assumptions versus data reality.	• Model data with probability functions (discrete and continuous) and interpret results. • Critically reflect on mathematical assumptions versus data reality.

Assessment Requirements

Applied problem-solving activity is used to evaluate student understanding of the underpinning mathematical concepts. This activity is undertaken as it is likely students will have to apply the mathematical knowledge on paper while trying to solve a particular problem before applying the processes computationally.

Each student must choose their own dataset (**minimum 300 rows, 5+ variables**). The dataset **must not** be from Kaggle/UCI pre-cleaned sources — instead, it should come from either (a) personal/professional interest, (b) government/NGO open data portals, or (c) raw CSV/web scraping.

Two main challenges in this coursework are:

- Handling a considerably large dataset by performing statistical and mathematical operations on it.
- Selecting attributes from the dataset to satisfy self-defined criteria.

Marking Criteria

Assessment Weighting (100 marks)

- Dataset choice & justification – 10
- Vector & matrix operations – 20
- Descriptive statistics & EDA – 20
- Probability theory & random variables – 20
- Probability distributions (PMF, PDF) – 20
- Critical reflection – 10

Assessment Success Guidance

Task 1 – Dataset Selection (10 marks)

- Each student must **choose their own dataset** (minimum 300 rows, 5+ variables).
- The dataset must not be from Kaggle/UCI pre-cleaned sources — instead, it should come from either (a) personal/professional interest, (b) government/NGO open data portals, or (c) raw CSV/web scraping.
- Submit a 1-page explanation of data provenance, variables, and why it is meaningful to you.

Task 2 – Mathematical Application

Perform the following analyses on your dataset (showing all working, derivations, and explanations).

1. **Linear Algebra – Vectors (5 marks)**
   - Represent at least one variable as a vector.
   - Perform at least two vector operations (dot product, norm, cosine similarity) and interpret the result in context of your dataset.
2. **Linear Algebra – Matrices (15 marks)**
   - Construct a data matrix **X**.
   - Compute $\mathbf{X}^T\mathbf{X}$ and interpret what it represents in your dataset.
   - Perform a rank analysis or determinant-based interpretation and explain what it means in practice.
3. **Descriptive Statistics & EDA (20 marks)**
   - Compute summary statistics (mean, variance, skewness, kurtosis) for two key variables.
   - Produce at least two visualisations (histogram, scatterplot, boxplot) and mathematically interpret the shape/patterns.
4. **Random Variables & Probability (20 marks)**
   - Choose one discrete and one continuous variable from your dataset.
   - Model them as random variables and write down their PMF (discrete) and PDF (continuous).
   - Justify mathematically why these distributions are (or are not) appropriate.
5. **Probability Functions – Discrete & Continuous (20 marks)**
   - Fit one standard discrete distribution (e.g. Binomial, Poisson) to part of your dataset.
   - Fit one standard continuous distribution (e.g. Normal, Exponential) to another variable.
   - Compare theoretical vs empirical fit and discuss limitations.

Task 3 - Critical Reflection (10 marks)

- Where did the mathematics align with your dataset, and where did it break down?
- Which assumptions (independence, normality, randomness) were unrealistic, and what was the impact?
- How could misuse of these mathematical tools mislead a data scientist?

Assessment Guidance Support and Formative Feedback

All the following must be submitted within the due date to be considered for a complete submission for this assessment.

(1)Report - your report should contain

The report should use 12-point font in Times New Roman, 1-inch margins, and double line spaced. The report should be properly paged, paragraphed, and sectioned, and include the following sections in order in a (.pdf) file.

I. Cover Page
II. Table of Content

III. Tasks:
Task:1 Submit a 1-page explanation of data provenance, variables, and why it is meaningful to you.

Task:2 Complete all 5 parts (max. 1000 words) along with necessary screenshot(s) (excluded from word count)

Task:3 Critical reflection (max. 250 words) along with necessary Screenshot(s) (excluded from word count)

IV. References:
Referencing any point in your discussion section(s) If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

(2) Interactive Python File

The supporting file you submit with your report should be an interactive python file (**.ipynb** format), contains codes you implemented with proper titles (in text blocks) and comments (in code blocks) and the file name should be "your name" with ipynb extension, like **"your name.ipynb" (E.g. *michaelng.ipynb*).**

If you are unable to upload the "your name.ipynb" file directly through your submission box, you can alternatively, upload a "zipped" or, "rar" folder that contains your .ipynb file in it.

Additional Information (if required):

- You will need to review **Weeks 1 to 8** Seminars and Lab activities to complete these tasks.

- Remember, you MUST perform **all 3 tasks** in **python** using appropriate libraries e.g., NumPy, SciPy, Pandas etc. However, for preliminary preprocessing steps (if necessary) you may use excel only to some extent.

Assessment Criteria (Grade Boundaries or Rubric):

Criteria	Excellent	Satisfactory	Not Satisfactory	Not Attempted
Understanding Data, The understanding and interpretation of data	Understanding about the data is clear and appropriate for all tasks	Understanding about the data is good but not discussed in sufficient detail in some tasks	Understanding and interpretation of data is somewhat clear, and/or is not discussed in sufficient detail for most of the tasks	No understanding is reflected
Appropriateness of Discussion about the Data and the chosen/proposed a method(s)	Understanding about chosen method is appropriate and discussed accurately in the report	Understanding about the chosen method is ok but could be discussed better	Understanding and discussion about the method is somewhat ok but is lacking sufficient detail	The choice of method is not appropriate and/ or, not discussed at all
Execution How smoothly does the Codes Execute- any error there?	Codes for all tasks executed correctly	At least 3 task solutions executed correctly	At least 2 task solutions executed correctly	No codes provided or, most have syntax error
Code Quality and Commenting structures of codes used during the implementation	Code Quality is appropriate and commented properly	Code Quality Needs work and commenting is missing in some places	Code solutions do not cover the entirety of the problem definition and/or, not commented appropriately	No codes provided or, code provided is very poorly formatted and no comment provided
Quality of Information and/or, supporting Files ability to provide details or supporting documents to support the report.	The supplied ipynb file contains all of the code and text blocks to appropriately implement and explain the solutions to the tasks	The ipynb file contains all of the coding solutions for the tasks but some text blocks are missing/explanation provided are not sufficient	The ipynb file does not have all of the coding solutions for the tasks and some/all text blocks are missing/explanation provided are not sufficient	An ipynb file was submitted but it did not open/did not have any code and/or text blocks or no ipynb file was submitted
Report Submission ability to provide a submission that meets the requirements given.	Correct files submitted only with no issues	Correct files submitted only but some issues with files provided	Correct files submitted but with unwanted files and possibly issues in files provided	Incorrect/no files were submitted.

Contact for Queries/ who you can contact for further information or queries

Should you have any question, please do not hesitate to contact Dr Michael Ng at michael.ng@roehampton.ac.uk

Ethical Requirements

Academic Misconduct:

"Academic integrity and honesty are fundamental to the academic work you produce at the University of Roehampton. You are expected to complete coursework which is your own and which is referenced appropriately. The university has in place measures to detect academic dishonesty in all its forms. If you are found to be cheating or attempting to gain an unfair advantage over other students in any way, this is considered academic misconduct, and you will be penalised accordingly."

Further details about "Student Code of Conduct" and "Disciplinary Regulations" can be found at:

https://www.roehampton.ac.uk/corporate-information/policies/

Use of Artificial Intelligence (AI)

The assessment is designed so that the use of AI during the assessment is irrelevant or impossible. You are welcome to use AI to support learning and preparation for this assessment. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions. You can find the guidance on the use of AI here*.

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

**You can find the student guidance on the use of AI in the Library and Nest.*

Referencing

All required references are available on Moodle.

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who

can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 50%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?